SCALABILITY & FUTURE PLAN

Date	01 October 2026
Team ID	SWTID-2026-2430
Project Name	Team Enrollment Confirmation – FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current FitBuddy project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current prototype state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	**Local SQLite Database & File Storage**	SQLite restricts high concurrent write transactions, causing potential performance bottlenecks and database locks under heavy concurrent user traffic.	**High**
2	**Synchronous Gemini API Requests**	Direct API calls during request execution can cause high latency or page timeout for users during peak network congestion or model delays.	**High**
3	**Basic In-Memory State & Single Worker Deployment**	Single instance deployment lacks auto-scaling capability and failover resilience, limiting system availability under spikes in user activity.	**Medium**
4	**Lack of Multi-Tenant User Authentication**	Current session tracking is basic, requiring robust OAuth2 / JWT authentication before scaling to public production environments.	**Medium**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Single instance FastAPI app handling sequential user submissions.	Deploy with Gunicorn/Uvicorn workers behind an NGINX load balancer; containerize using Docker & Kubernetes for auto-scaling.
2	**Data Storage**	Local SQLite database storing workout logs and user feedback.	Migrate to Managed PostgreSQL for relational persistence and Redis Cache for instant retrieval of frequently accessed fitness plans.
3	**Performance**	Direct REST calls to Gemini API causing response delay.	Implement asynchronous task queues (Celery / RabbitMQ) for background plan processing and webhooks for instant client notification.
4	**Security**	Basic form validation and environment variable API keys.	Integrate OAuth2 with JWT authentication, rate limiting (FastAPI-Limiter), HTTPS encryption, and AWS Secrets Manager key storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1		Month 1 - Month 2	

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
	Mobile-Responsive UI refinement, PDF exports for workout plans, and OAuth login integration.		Improved user retention and seamless user onboarding.
Phase 2	Wearable device integration (Apple Health, Fitbit API) for auto-syncing actual workouts and calorie burn.	Month 3 - Month 4	Hyper-personalized workout adjustments based on live biometrics.
Phase 3	Computer vision exercise form correction via smartphone camera and multi-language support.	Month 5 - Month 6	Enhanced safety, interactive experience, and global reach.
Phase 4	AI Community Coaching Hub, peer challenges, social sharing, and personalized meal recipe generation.	Month 7 - Month 9	Complete fitness ecosystem driving user engagement and growth.

Team Work Allocation & Responsibilities:

S.No	Team Member	Assigned Responsibility	Expected Contribution
1	**Divya.D**	Requirements & Coordination	Define user inputs, fitness goals, workflows, and requirements; coordinate and consolidate project documentation.
2	**Deepika.M**	Gemini AI Integration & Prompts	Design prompts for structured 7-day workout plans, nutrition/recovery tips, and feedback-based plan revisions.
3	**Dharani.T**	FastAPI Backend & Routing	Implement form handling and routes for plan generation and feedback; connect backend routes to AI functions.
4	**Divya shree.R**	Frontend & User Experience	Build HTML/Jinja2 pages for inputs, plans, tips, and feedback; style responsive, modern layouts across devices.
5	**Divya Dharshini.V**	Database, Testing & Deployment	Support SQLite/SQLAlchemy plan history, execute end-to-end workflow testing, and verify deployment configuration.